

Jenkins Pipeline Steps

Understanding Built-in Jenkins Pipeline Functions

Topics

Git

Checkout

Shell Commands

Workspace

File Operations

Build Information

Archive

Stash

Utility Steps

Git Step

Clone a repository

```
git url: 'https://github.com/company/app.git'
```

Options

- branch
- credentialsId
- url

Explain

When to use **git** vs **checkout**

Checkout Step

More flexible than git.

Example

```
checkout scm
```

Advanced checkout

```
checkout([ $class: 'GitSCM' ... ])
```

Explain

Multibranch Pipelines

pwd()

Display current workspace.

```
pwd()
```

Output

```
/var/lib/jenkins/workspace/demo
```

dir()

Change directory

```
dir("src"){ sh "ls" }
```

Useful for

Mono Repo

Nested folders

deleteDir()

Delete workspace

```
deleteDir()
```

Benefits

Clean Build

Prevent old files

cleanWs()

Workspace Cleanup Plugin

```
cleanWs()
```

```
deleteDir()
```

```
cleanWs()
```

Difference

writeFile()

Create files

```
writeFile( file:'version.txt', text:'Version 2.0' )
```

readFile()

Read files

```
def version=readFile('version.txt') echo version
```

fileExists()

Verify

```
if(fileExists("pom.xml")){ echo "Found" }
```

archiveArtifacts()

Archive build output

```
archiveArtifacts '*.jar'
```

Explain

Artifacts

Reports

Downloads

stash()

Store temporary files

```
stash name:'build', includes:'target/'
```

unstash()

Retrieve files

```
unstash 'build'
```

Explain

Multi-agent Pipelines

currentBuild

Useful properties

```
currentBuild.number currentBuild.result currentBuild.duration
```

Build Status

Display

SUCCESS

FAILURE

UNSTABLE

ABORTED

NOT_BUILT

Explain

When each occurs.

Utility Options

`timestamps()`

`ansiColor()`

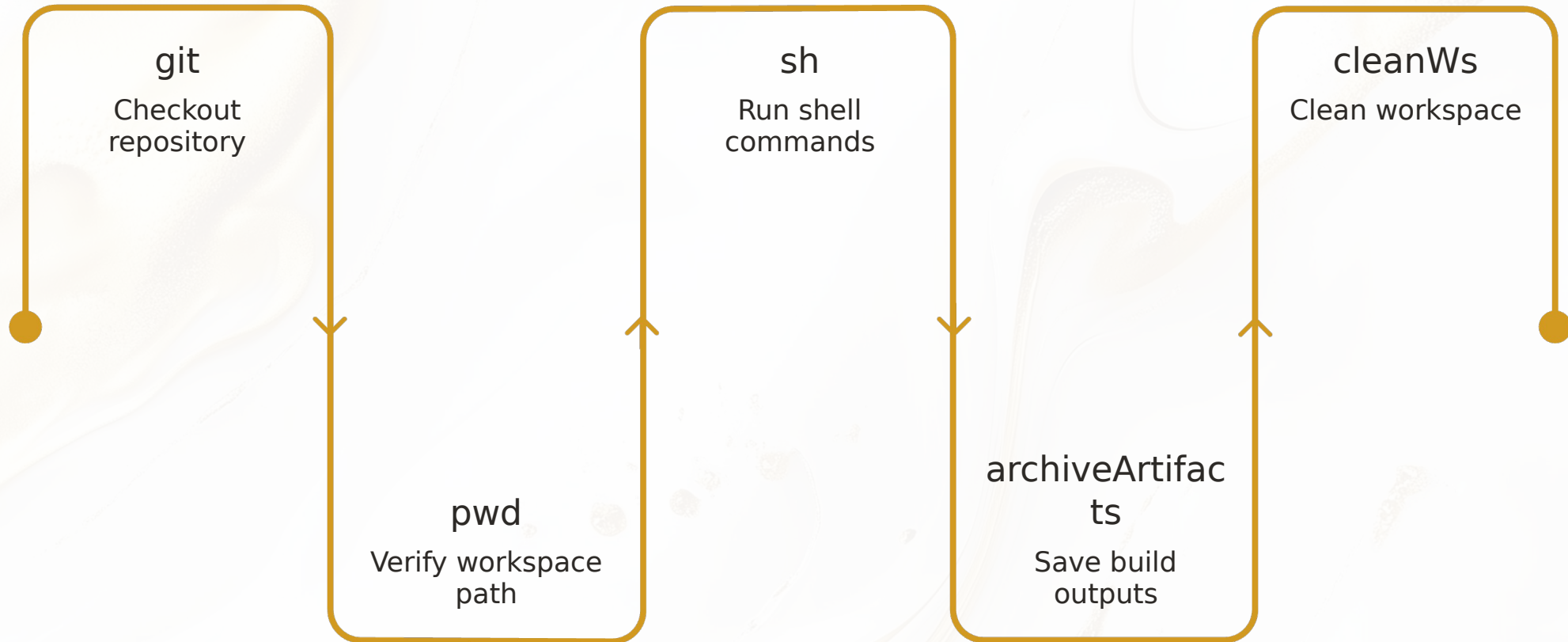
`disableConcurrentBuilds()`

`buildDiscarder()`

`skipDefaultCheckout()`

Combining Multiple Steps

Example



Real Pipeline

Enterprise Pipeline Example

